IT ROOTS
EMPOWERING MINDS

#Empowering Minds

Rooted | Recruited

ROOTED IN SKILLS, RECRUITED FOR SUCCESS.

Software Testing with AI

TRANSFORMING POTENTIAL INTO PROFESSIONALS.

www.itroots.co.in

About ITROOTS

At ITROOTS, we believe in transforming potential into professional excellence.
With over two years of proven success in Online training, ITROOTS now brings its industry-focused expertise OFFLINE , empowering learners across India to become job-ready professionals.

Our mission is to equip every learner with industry-ready skills, hands-on experience, and career confidence through AI-integrated, project-based learning.

Why Choose ITROOTS?

Choosing the right training partner can define your career and at ITROOTS, we turn your goals into guaranteed outcomes.

Why learners and employers trust us:

100% Placement Guarantee: We ensure every learner achieves their career goal.

Refund Guarantee: If we don't deliver as promised, your investment is safe.

Post-Placement Support: Continued guidance even after you're hired.

Lifetime Career Assistance: Ongoing mentorship and growth support

Project-Oriented, AI-Integrated Training: Learn through live, industry-level projects.

Expert Mentorship: Trainers with 10+ years of industry experience

Daily 5–6 Hours of Practical Training: Focused, hands-on sessions.

Regular Assessments & Mock Interviews: Build confidence before stepping into interviews.

At ITROOTS, you don't just learn, you build, apply, and grow with every step.

Program Overview

The Software Testing with AI Program is a complete 6-month hands-on journey designed to make you an industry-ready QA professional.

You'll master manual and automation testing, Selenium, API testing, and AI-based tools that make software testing faster, smarter, and more reliable for real-world applications.

Duration: 6 Months **Mode :** Online / Offline

Ideal for: Graduates, Job Seekers, and Working Professionals

The Learning Journey

Every learner at ITROOTS follows a path that ensures real progress and measurable results:

Enroll & Set Goals

Begin your transformation with a clear career roadmap.

Learn Intensively

Master Software Testing with AI through focused, daily sessions.

Build & Innovate

Work on real, AI-integrated projects that sharpen your skills.

Test & Improve

Strengthen your performance with assessments and mock interviews.

Get Placed

Land your dream job with our 100% placement guarantee.

Keep Growing

Enjoy lifetime mentorship and career guidance - always.

Syllabus

Introdution

IT Basics | SDLC Introduction | Types of SDLC Models – Waterfall, Spiral, Prototype, V-Model, Hybrid | Roles & Responsibilities of IT Resources | SQL – DDL, DML, DCL, Constraints, Functions | Soft Skills – Communication Skills, Group Discussion, Writing Skills, Interview Skills, Resume Writing | Types of Testing – Functional, Non-Functional, Smoke, Sanity, Regression | Types of Requirements | Test Cases & Scenarios | Test Case Design Techniques | STLC – Requirement → Planning → Design → Execution → Reporting | Test Management Tools | Test Plan | Defect Tracking | ISTQB Foundation Topics | Mobile Application Testing | Weekly Test + Q&A |

Java Basics

Syntax, Data Types, Variables, Operators | Functions/Methods | Static & Non-Static | Constructors | IIB/SIB | OOP – Class, Object, Inheritance, Polymorphism, Abstraction, Encapsulation | String Class | Collections Framework | Exception Handling | Multithreading | Weekly Java Practice + Q&A |

Agile & Scrum

User Stories | Story Points | Planning Poker | Product Backlog | Sprint Backlog | Scrum & Scrum Team | Standup Meetings | Sprint Planning | Role of Product Owner | Role of Scrum Master | Team Size Logic (6–9) | Sprint Duration Logic (2–4 weeks) | Sprint Retrospective | Burndown & Burnup Charts | Kanban | Advantages & Disadvantages | Weekly Agile Activities + Q&A |

Automation Basics

When to Automate | Automation Tools Landscape | Selenium – JAR Setup, WebDriver Architecture | HTML Tags & Attributes | Locators – ID, Name, XPath, CSS, LinkText | Automating Buttons, Links, Text Fields, Radio Buttons, Checkboxes, Dropdowns, Search Boxes | Cross-Browser Launching & Navigation | Close vs Quit | Synchronization – Implicit/Explicit Waits | Multiple Windows Handling | Screenshots | File Upload Automation | Robot Class | Clipboard Handling | Select Class | Browser Navigation | Data-Driven Testing with Excel + Apache POI | Weekly Selenium Tasks + Q&A |

Syllabus

Jenkins

Importance in Frameworks | Installation & Configuration | Jenkins Workspace | TestNG Reporting with Jenkins | POM.xml in Jenkins | Continuous Integration Basics | Writing Test Plans | Requirement Understanding & Scenario Identification | Writing/Reviewing/Executing Test Cases in QC | Bug Reporting in Bugzilla | Smoke/Functional/Integration/System Testing | Adhoc & Exploratory Testing | Security Testing | STCM | Traceability Matrix | Monthly Status Reporting | SQL Backend Validation | Daily Standups | Burndown Chart Analysis | Weekly CI/CD Practice + Q&A |

API Testing

Postman (Requests, Headers, Params, Body, Assertions) | REST Assured (Setup, Serialization, Deserialization, Validations) | API Automation | Performance Testing – JMeter (Thread Groups, Samplers, Listeners) | LoadRunner Basics | Weekly API + Performance Tasks + Q&A |

Framework Development

Properties File | JUnit – Annotations, Checkpoints, Parameterization, Build.xml, Batch Execution, HTML Reports | TestNG – Annotations, Checkpoints, Data Provider, XML Config, Parallel Execution, Include/Exclude, Listeners Interface | Selenium Grid | AutoIT | Log4j | Maven – Importance, Terminologies, POM.xml, Dependencies, TestNG Integration | Weekly Framework Building + Q&A |

Aptitude Training

Number System | Percentages | Profit & Loss | Simple Interest | Time & Work | Time, Speed & Distance | Permutation & Combination | Probability | Geometry | Number Series | Coding-Decoding | Directions | Blood Relations | Puzzles | Cube & Dice | Set Theory | Weekly Aptitude Practice |

Software Testing Module

MODULE -1

IT Basics

- What is SDLC?
- Types of SDLC Models
- Waterfall, Spiral, Prototype, V-model, Hybrid Derived Roles and responsibilities of IT resources
- JAVA BASICS
- Introduction to Java o Data Types, Variables, Strings, Operators Control & Looping statements o OOP's – Class, Object, Inheritance, Polymorphism, Abstraction, Encapsulation
- String class, Collection Framework o Exception Handling, Multithreading
- Data Types, Functions/methods, static/Non-static, Operators, Constructors, IIB/SIB

Manual Testing

- Types of Testing
- Types of requirements
- TEST CASE AND SCENERIOS
- Test case design techniques
- STLC
- Test Management Tool
- Test Plan • Defect Tracking
- Projects • ISTQB Foundation
- Introduction to Mobile Application Testing
- Resume Building
- HR Classes

Agile

- Understanding user stories
- How to give user story points
- What should be the format of user stories?
- How we do planning poker activity
- Explanation on product back log
- Explanation on Sprint backlog
- What is scrum?
- Explanation on Scrum team
- Purpose of standup meetings
- What we do in sprint planning
- Role of product owner
- Role of Scrum master
- Explanation on why team size should be between 6-9
- Explanation on why sprint duration should be 2 – 4weeks.
- Sprint retrospect meetings
- What are burn down and burn up charts?
- Why KANBAN charts are used.
- Advantages & Dis-Advantages

SQL

- Syntax
- Datatypes, Operators
- DDL, DML, DCL
- CONSTRAINTS, FUNCTIONS

SOFTSKILL

- Communication skills
- Group discussion
- Writing skills
- Interview skills
- Resume writing

MODULE-2

SELENIUM

- What is automation?
- Different automation tools available in market
- When do go for automation?
- How to configure selenium jars in eclipse.
- Architecture of selenium WebDriver
- Explanation on various HTML tags and attributes
- Different types of locators in Selenium
- Automating several fields of web application live online.
- Button
- Link
- Text field
- Radio button
- Checkbox
- Dropdown box
- Searchbox
- Launching and interacting different browsers in selenium WebDriver Different ways to perform navigation in selenium WebDriver.
- Difference between close and quit.
- Performing synchronization in selenium
- Handling multiple windows in selenium
- Taking screen shots in selenium
- Automating file upload
- Explanation on Robot class
- String selection class • Clipboard
- Synchronization practical usage while uploading file
- Automating select boxes using select class
- Navigating options in a browser
- Performing data driven testing in selenium by storing data on excel file using POI API

Software Testing Module

- Properties file
- What is Junit?
- Annotations of Junit
- Developing check points with Junit and Selenium
- Parameterizing test using Junit.
- Developing Build.xml
- Running test in batch
- Generating HTML based reports

TestNG

- Annotations of TestNG
- Developing check points with TestNg and Selenium
- Verifying page tile
- Verifying page text
- Verifying edit box
- Verifying links
- Verifying buttons and many more
- Parameterizing test using Junit.
- Importance of xml file in TestNG configuration
- Data provider Annotation
- Parameterizing test cases
- Listeners interface in TestNG Framework
- Running tests in parallel and generating reports
- Include and exclude mechanism to control test cases.
- Selenium Grid
- Auto IT
- Log4j

Maven

- Importance of Maven in FW development
- Understanding terminologies in Maven
- Creating Maven project and exporting in Eclipse
- Understanding POM.xml file and its dependencies
- Integration of TestNG with maven

Aptitude

- Number system, percentages, profit & Loss, simple interest
- Time And Work, Time Speed and Distance, Permutation and Combination, Probability, Geometry
- Number Series, Coding Decoding, Directions, Blood Relations, Puzzle Based Problems, Cube & Dice, Set Theory etc.

MODULE-3

Jenkins

- Importance of Jenkins in Test Frameworks
- Install and configure Jenkins.
- Configuring Jenkins settings and workspace
- TestNG Reporting plugin into Jenkins jobs.
- POM xml file code

Project

- Project execution work
- Write Test Plan
- Understanding requirement and identifying scenarios
- Write / review / excuse Testing Cases in Quality Centre
- Reporting defect in Bugzilla
- Performing smoke testing
- Performing functional, integration and system testing
- Performing adhoc testing, exploratory testing etc.
- Performing security testing
- Preparing STCM
- Preparing traceability matrix
- Preparing monthly status reports
- Validating data in the back-end database using SQL
- Attending daily stand-up meetings.
- Analyzing burn down chart

MODULE-4

API And Performance Testing

- POSTMAN
- RESTASSURED
- JMETER
- LOADRUNNER

MODULE-5

2 Projects

- Student Management System
- Product Inventory Management System

Real World Project : Insurance

Loan Granting System

1.Project Scope

The main objective of this project is to create a secured, robust loan management system. Here, all the information on Loan is available to every user 24*7.

2.Landing Page

Home Page
About Us Page
Contact Us

3.Account

Login Page Sign Up Page

4.Loan Application

Normal User can apply for loan and also can view applied loan status Admin can assign loan officer and field officer for particular application Field officer can do cross check of all documents and make remark Loan officer can review loan application can approve/reject Loan application

5.Admin

Admin can add/modify/delete user
Admin can add/modify/delete loan officer
Admin can add/modify/delete filed officer
Admin can generate all kind of reports

Insurance Management System

1.Project Scope

The main objective is to build a secured, robust insurance management system where the policies are managed properly. All the information of user, their policy, category of insurance is maintained on a single platform.

2.Landing Page

Home Page
Category Page
Sub-Category Page
Policy Page

3.Account

Login Page
Sign Up Page

4.Category

Admin can add/modify/delete category
Normal User can view and get details of categories

5.Sub-Category

Admin can add/modify/delete sub-category Normal User can view and get details of sub- categories

6.Policies

Admin can add/modify/delete Policy
Normal User can view and get details of Policy Normal User can apply for policy
Admin can approve/reject policy request
User can see details of all enrolled policies

HTML CSS JS Java 17 MySQL spring

Our Role in Your Success

At ITROOTS, our commitment goes beyond teaching. We partner with you throughout your career journey. From learning the latest technologies to securing your dream job, we make sure you're guided, supported, and ready for real-world success.

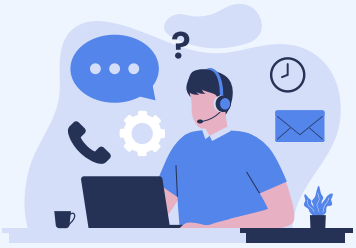
1. Skill Advancement & Technology Mastery

Our course builds your expertise in ensuring software quality through manual and automation testing. You'll gain practical knowledge of tools and frameworks that make testing smarter and faster.

Technologies You'll Master:

- Manual Testing | Selenium
- API Testing
- JUnit
- TestNG
- Automation Frameworks
- Performance Testing Tools
- Bug Tracking Systems

2. Continuous Career Support



We don't just train you; we prepare you to thrive.
Our dedicated career team provides:

- Resume & LinkedIn optimization
- Technical & HR mock interviews
- Personalized career counselling
- Access to exclusive job networks and hiring partners
- Lifetime mentorship for continuous growth and upskilling

3. Guaranteed Placement Assistance

Your success is our goal.
Through our Hire → Train → Deploy model, we ensure every learner is:

- Trained with AI-integrated, project-based sessions
- Evaluated through tests and mock interviews
- Deployed into top IT companies with complete career readiness

With our 100% Placement Guarantee and Refund Assurance, your journey with ITROOTS is result-driven, transparent, and focused on success.

Alumni Success Stories

ITROOTS is one of the best institutes for training and placement. The trainers are experienced, supportive, and always approachable. I completed my Full Stack Java Certification from ITROOTS and got placed as a Software Developer. Grateful to the entire ITROOTS team for their constant guidance.
~ **Priya Wagh**

Joining ITROOTS was the best decision for my IT career. In just six months, I learned Frontend, Backend, and Database technologies and got placed within a week after completing the course. The placement team is truly dedicated and helpful throughout the process.
~ Harsh Patil

ITROOTS is an excellent place for learning and placement. The team is very supportive and helps you build knowledge from the basics, even if you come from a non-IT background. Thank you, ITROOTS, for helping me start my career with confidence.
- Mayur Aware

250+ Success stories so far and counting
visit our website for more success stories and get inspired

Our Client Partners

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and many more....

Our Commitment

"Rooted in Skills. Recruited for Success."

At ITROOTS, we promise outcomes, not just education. With 100% Placement Guarantee, Refund Assurance, and Project-Oriented Training, every learner becomes a professional with confidence and clarity. We prepare you for industry success with real skills, real projects, and lifelong support.

Your growth doesn't stop at placement, it starts there.

Contact Us

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